INDEX

NAME: ISHWARIYA R STD.: SEC.: ROLL NO.: SUB.: COMPUTER NETWORKS

AI&DS - III - YEAR

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"Wherever you see "cover", it means both the front and back."

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S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
11) b)		Routing Information Protocol		
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completed

```
Command Prompt

C:\Users\jagan>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

```
C:\Users\jagan>hostname
MSI
```

```
C:\Users\jagan>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c29:7edc:b765:225b:1dc6:645a
   Temporary IPv6 Address. . . . . . : 2401:4900:1c29:7edc:58bc:ac4a:e234:a299
   Link-local IPv6 Address . . . . . : fe80::9e7e:3ef5:9420:2b27%7
   IPv4 Address. . . . . . . . . . . : 192.168.1.7
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%7
                                       192.168.1.1
```

EX NO:1
DATE: 29/7/26

NETWORK

AIM:

To study

In linux and

COMMANDS:

* arp -a →

along u

addres

* hostname →

* ip Config →

* netstat →

a

* nslookup

DI

* pathping

addres

of each

EX NO:1
DATE: 24/7/26

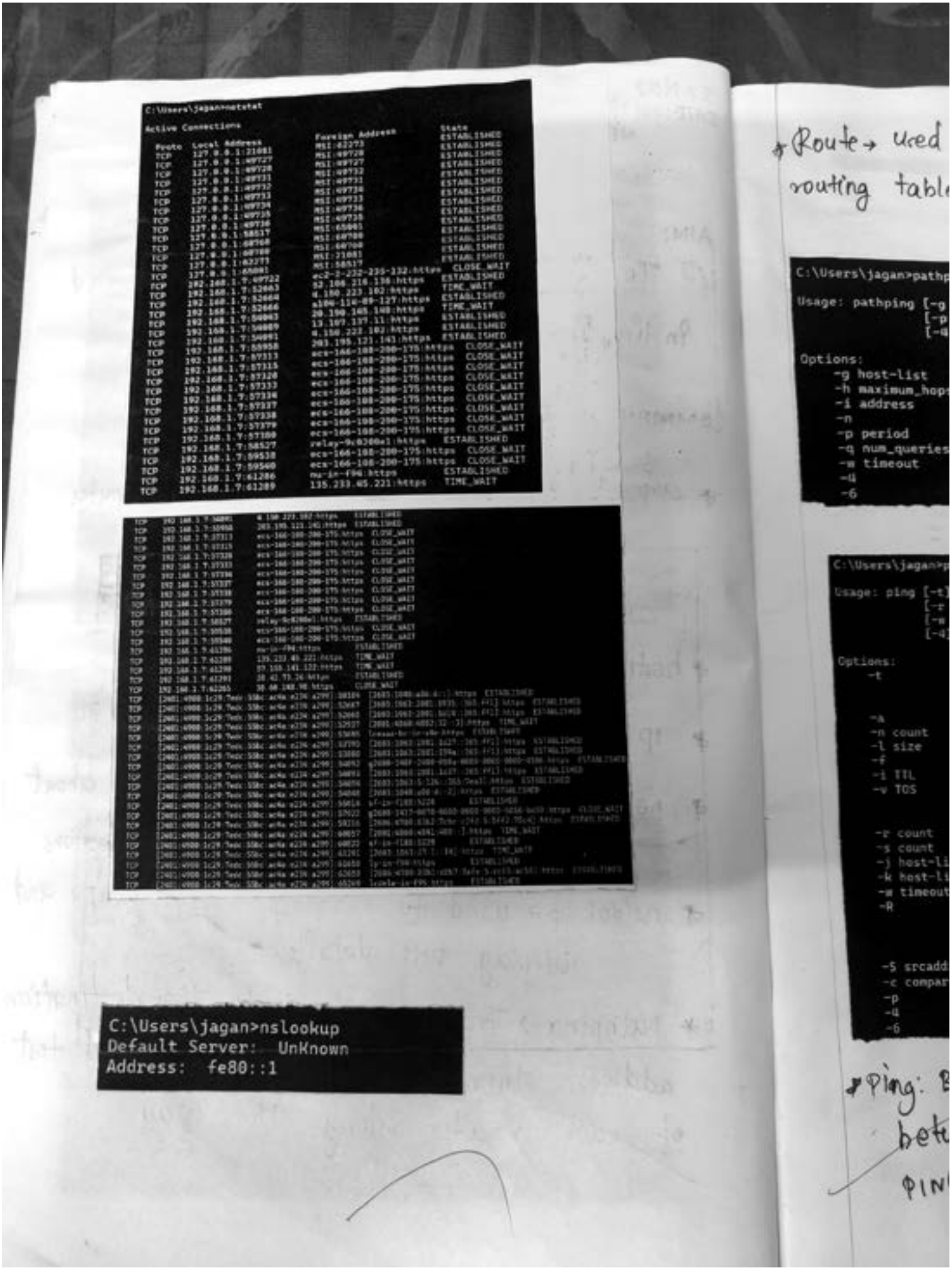
NETWORK COMMANDS USED IN LINUX

AIM:

To study various Network commands used in linux and Windows.

COMMANDS:

* arp -a → shows IP address of the computer along with the IP address and mac address of your router
* hostname → Displays name of the Computer
* ip Config → Displays Configuration in the PC
* netstat → Displays a variety of statistics about a computer's active TCP/IP Connections.
* nslookup → used to perform DNS lookups and Display DNS details.
* pathping → Tracks the route to the destination address then launches a 25 second test of each router along the way



C:\Users\jagan>netstat

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:21081	MSI:62273	ESTABLISHED
TCP	127.0.0.1:49727	MSI:49728	ESTABLISHED
TCP	127.0.0.1:49728	MSI:49727	ESTABLISHED
TCP	127.0.0.1:49731	MSI:49732	ESTABLISHED
TCP	127.0.0.1:49732	MSI:49731	ESTABLISHED
TCP	127.0.0.1:49733	MSI:49734	ESTABLISHED
TCP	127.0.0.1:49734	MSI:49733	ESTABLISHED
TCP	127.0.0.1:49735	MSI:49736	ESTABLISHED
TCP	127.0.0.1:49736	MSI:49735	ESTABLISHED
TCP	127.0.0.1:58517	MSI:65003	ESTABLISHED
TCP	127.0.0.1:60760	MSI:60780	ESTABLISHED
TCP	127.0.0.1:60780	MSI:60760	ESTABLISHED
TCP	127.0.0.1:62273	MSI:21081	ESTABLISHED
TCP	127.0.0.1:65001	MSI:58517	CLOSE_WAIT
TCP	192.168.1.7:49722	ec2-2-232-235-132:https	ESTABLISHED
TCP	192.168.1.7:52663	52.108.216.136:https	TIME_WAIT
TCP	192.168.1.7:52664	4.150.223.102:https	ESTABLISHED
TCP	192.168.1.7:52666	a104-114-89-127:https	TIME_WAIT
TCP	192.168.1.7:54063	20.190.165.148:https	ESTABLISHED
TCP	192.168.1.7:54089	13.107.137.11:https	ESTABLISHED
TCP	192.168.1.7:54091	4.150.223.102:https	ESTABLISHED
TCP	192.168.1.7:55958	203.195.121.143:https	ESTABLISHED
TCP	192.168.1.7:57313	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57315	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57328	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57333	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57336	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57337	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57338	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57379	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:57380	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:58527	relay-9c8300e1:https	ESTABLISHED
TCP	192.168.1.7:59538	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:59540	ecs-166-108-200-175:https	CLOSE_WAIT
TCP	192.168.1.7:61286	nu-in-f94:https	ESTABLISHED
TCP	192.168.1.7:61289	135.233.45.221:https	TIME_WAIT

C:\Users\jagan>nslookup
Default Server: UnKnown
Address: fe80::1

* Route → used
routing table

C:\Users\jagan>pathp

Usage: pathping [-g
[-p
[-q

Options:
-g host-list
-h maximum_hops
-i address
-n
-p period
-q num_queries
-w timeout
-4
-6

C:\Users\jagan>p

Usage: ping [-t]
[-a
[-n
[-l

Options:
-t

-a
-n count
-l size
-f
-i TTL
-v TOS

-r count
-s count
-j host-li
-k host-li
-w timeout
-R

-S srcadd
-c compar
-p
-4
-6

* Ping: B
bet
PIN

* Route → used to show / manipulate the IP routing table.

```
C:\Users\jagan>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

```
C:\Users\jagan>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

* Ping: Best way to find connectivity between two nodes.
  PING ICMP to communicate to other devices

```
C:\Users\jagan>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
C:\Users\jagan>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
               If not specified, it defaults to 255.255.255.255.
  gateway      Specifies gateway.
  interface    the interface number for the specified route.
  METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.
```

RESULT!



```
  METRIC      specifies the metric, ie. cost for the destination

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, 224*

Pattern match is only allowed in PRINT command
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*                  ... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
              destination^    ^mask      ^gateway     metric^    ^
                                                       Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

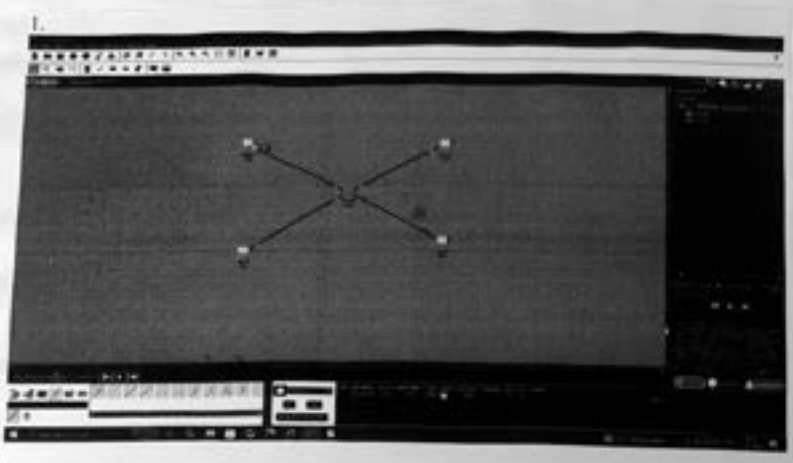
    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 1

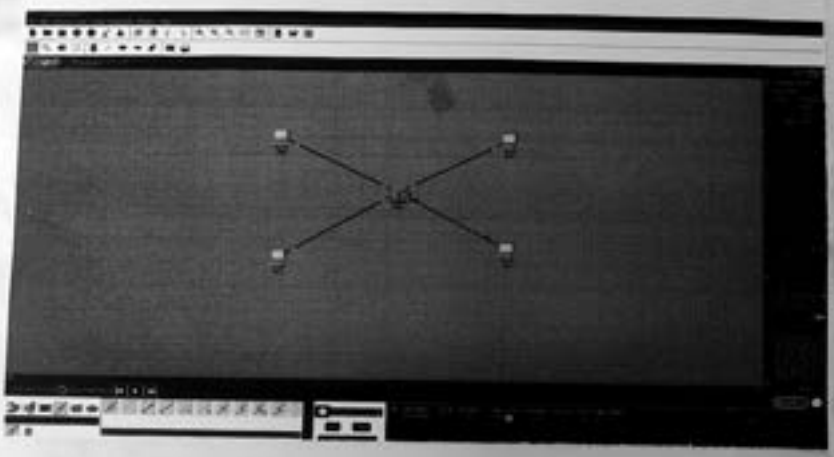
      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32
```

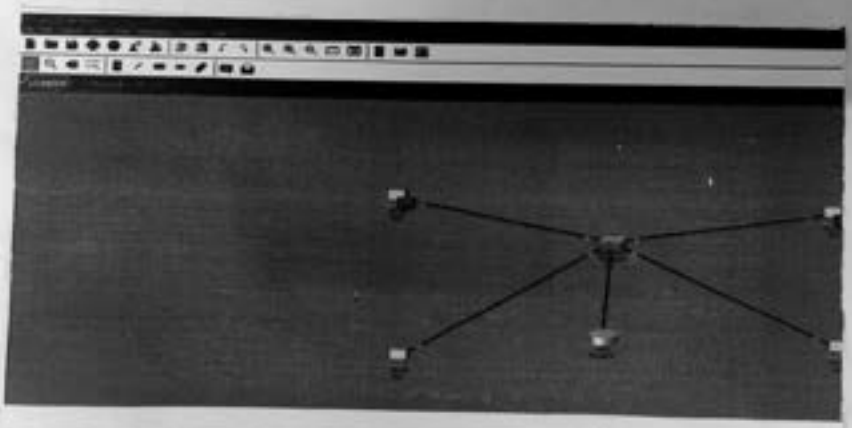
RESULT: Thus the network commands has been studied and executed Sucessfully

A) HUB TOPOLOGY:





B) ROUTER AND SWITCH TOPOLOGY:



Ex No:2
DATE: 25/7/25 NE

AIM: To
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Cisco
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Ex No:2
DATE: 25/7/25 NETWORK TOPOLOGY USING HUB, SWITCH AND ROUTER.

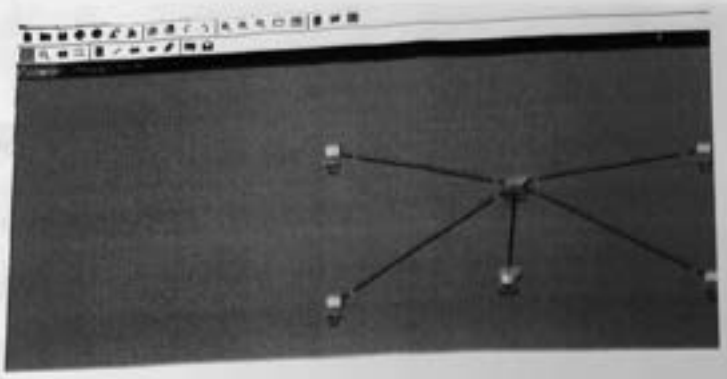
AIM: To create and simulate simple network topologies using a hub, switch & router in Cisco Packet Tracer and study data communication between devices.

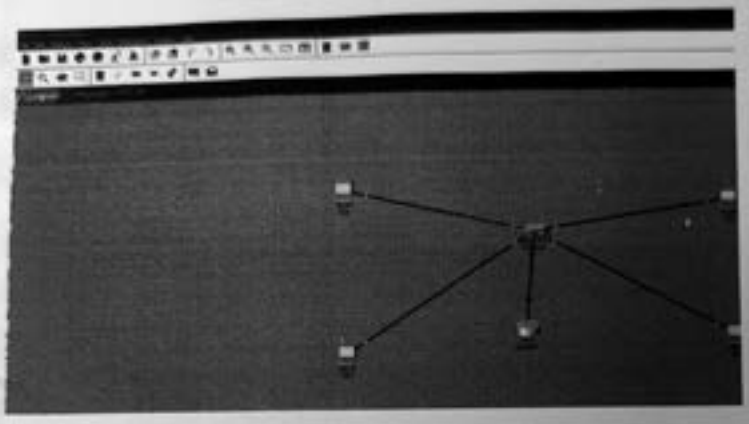
PROCEDURE:

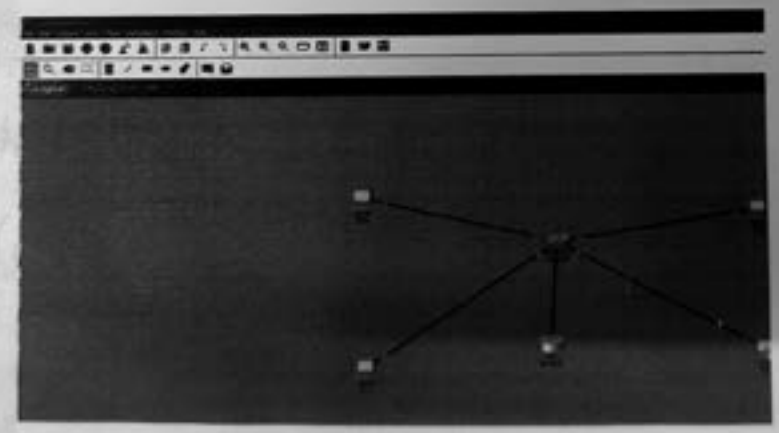
a) Hub Topology

1. opened Cisco packet Tracer
2. placed, Hub & 4PC in the workspace
3. Assigned IP addresses to all the 4PC's
4. Connected all the PC's to the Hub using copper straight-Through cables
5. Selected simple PDU (message icon) and clicked on the source PC and then double clicked on the destination PC
6. Selected the simulation mode and observed the packet transmission and communication between the PC's.

b) Switch and Packet Tracer.







1. Opened
2. placed
tow
3. Assigne
4. Conne
and
Zout
Cable
5. Sele
the
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obse
the

RESULT: T
Switch
created
was

1. Opened cisco packet tracer
2. placed one Router, one switch and four PC's in the workspace.
3. Assigned the IP address to all the 4 PC's
4. Connected all the 4 PCs to the switch and connected the switch to the Router using copper straight. Through cables.
5. Selected the simple PDU and clicked on the source PC and then double clicked on the destination PC.
6. selected the simulation mode and observed the packet transmission through the switch and Router.

RESULT: The network topologies using Hub, Switch and Router were successfully created in Cisco packet tracer and the PDU was transmitted successfully between the source and destination devices.

Difference between crossover cable and straight cable

int out the diagram below or have it handy as a reference

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B

: Next, strip approximately 1.5 cm of cable shielding from both ends. The ing tool has a round area to complete this task.

EX-3
1/8/2026 STUDY

Aim: study
Such as
Optic C

UTP:
Category
maximum
Tran
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Applica

SSTP ,
STP: C

maximu

Advant

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EX-3
1/8/2026

STUDY OF DIFFERENT TYPES OF NETWORK CABLES.

Aim: study of different types of Network cables. Such as UTP, STP, Coaxial Cable, Fibre Optic Cable.

UTP:

Category: Category 3, Category 5, category 5e

maximum data Transmission: 10bps, upto 100 mbps, 1Gbps

Advantages: cheaper in cost, Easy to install as they have a smaller overall diameter.

Disadvantages: more prone to (EMI) Electromagnetic interference and noise.

Application / Use: 10 Base T Ethernet, Fast Ethernet Grigabit Ethernet

SSTP, category 7

STP: Category: 6, 6a

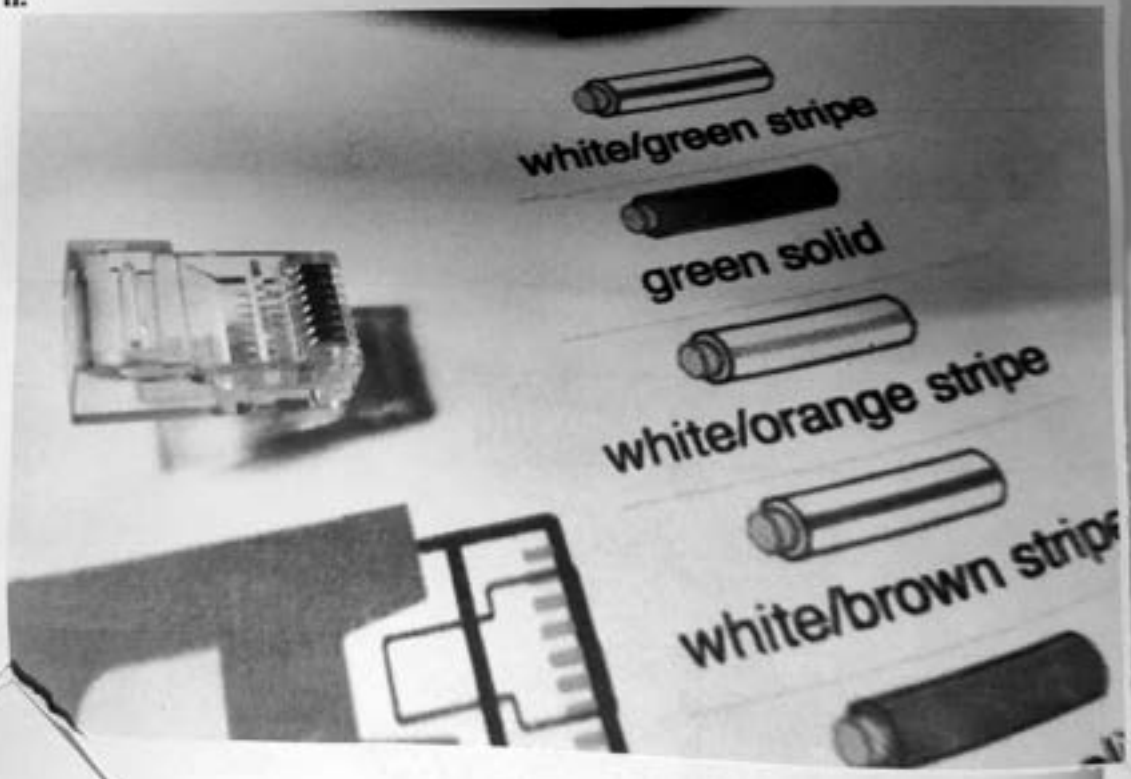
maximum data transmission: 10Gbps, 10Gbps

Advantages: Shielded, Fasta than UTP less susceptible to noise and Intefurence

Disadvantages: → Expensive
→ Greater Installation effort



4: Once the order is correct, bunch them together in a line, and if there are any
out farther than others, snip them back to create an even level. The difficult a
cing these into the RJ45 plug without messing up the order. To do so, hold the
the clip side facing away from you and have the gold pins facing toward you, a
n.



Application or Use:

Gigabit Ethernet, 10G Ethernet (55m)
widely used in data centres

Gigabit Ethernet, 10G Ethernet (100m)

Cable Type
Coaxial Cable

Category: RG-6, RG-59, RG-11

maximum Data Transmission: 10-100mbps

Advantages/Disadvantages: High bandwidth
Immune to interference
Low loss bandwidth
Versatile

→ limited distance,
Cost, size is bulky

Application/Use: speed of signal is 500m
Television network
High speed internet connections

category

fibre optics cable → single mode
multimode

Maximum data transmit - 100Gbps

Advantages - high speed, bandwidth, security

Disadvantages - Expensive. required skilled installer

Uses - maximum distance is 100M

b) make your own ethernet cross-over cable:

Tools needed

(i) Ethernet cabling
(ii) Crimping tool
(iii) Two RJ45 plugs
(iv) optional two plug shields.

Step 1: To start construction of the device, begin by threading onto the cable.

step 2: Next, strip approximately 1.5cm of cable sheeloling from both ends

The crimping tool has a round area to complete this task

step 3: Next, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. one end should be in arrangement A and the other in B.

step 4: once, the order is correct bunch them together in a line, and if there are any that stick out further from others, skip them back to create an even level.

step 5: Next, push the cable right in the notch at the end of the plug needs to be just over the cable shielding, and if it isn't that, means you stripped off two much shielding.

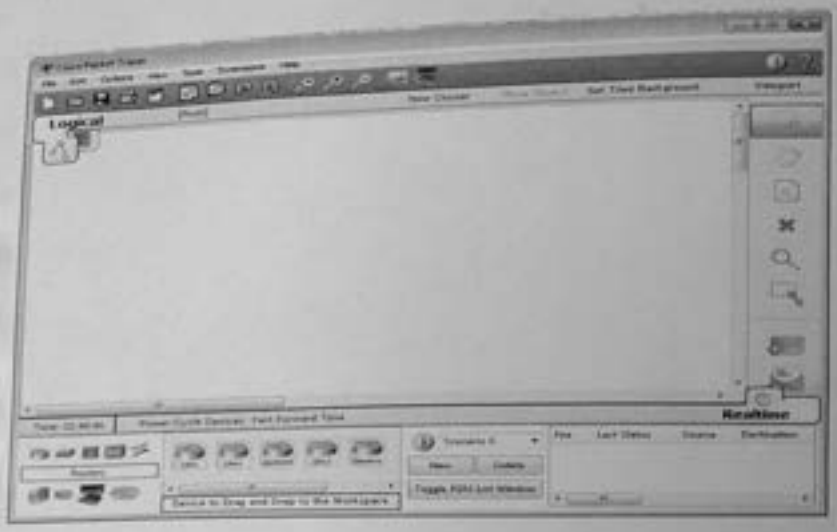
step 6: After the wires are seemely sitting inside the plug, insert it into crimping tool and push down.

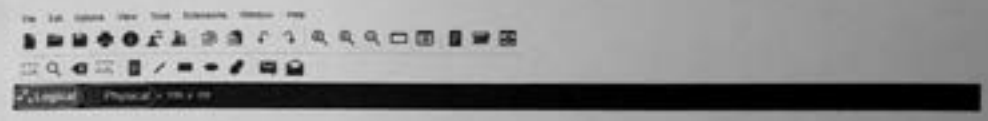
step 7: lastly, repeat for the other end using diagram B.

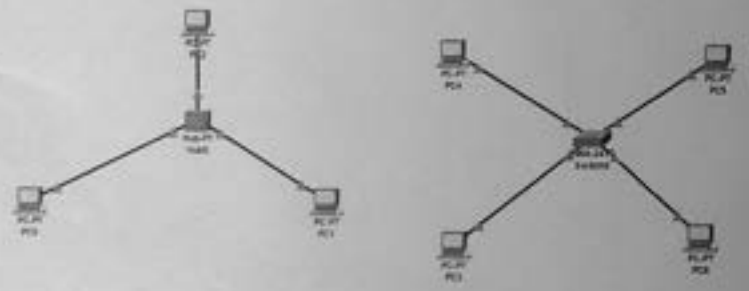
Result: Thus the different types of network cable has been studied along with their properties, Uses, Application.

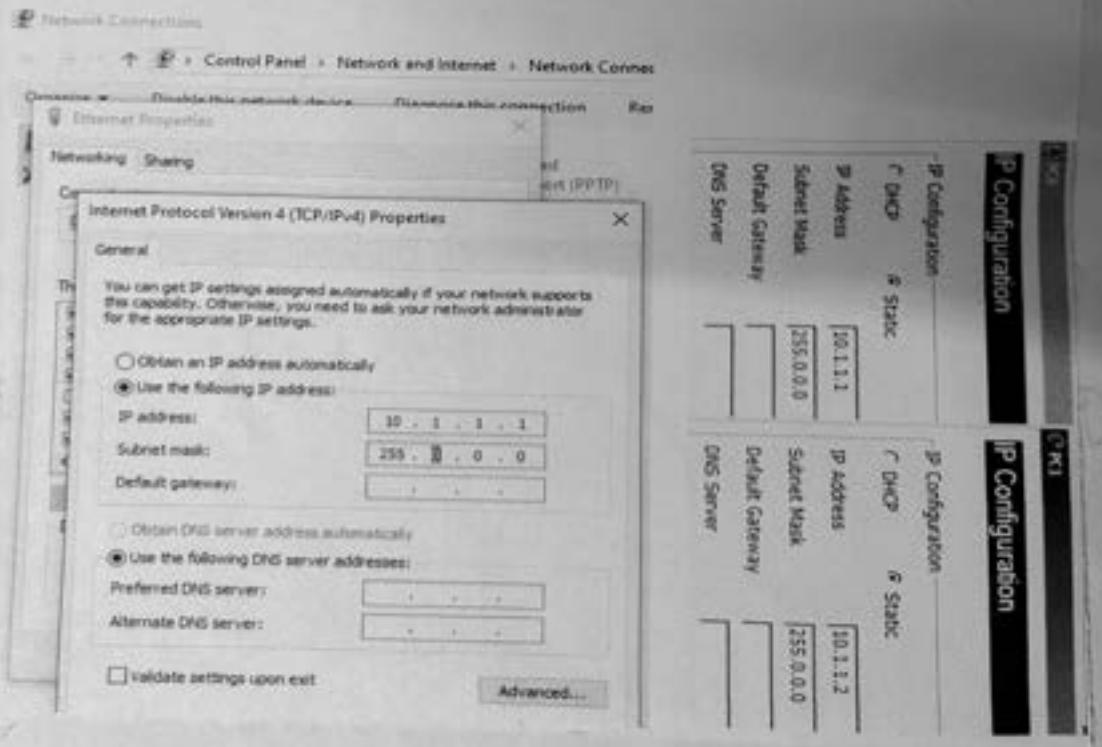
Ex-4
11/8/2026

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EX-4 CONFIGURATION OF LAN
11/8/2026

AIM: To set and configue a LAN (Local Area Network) using a switch and Ethernet cables in your lab.

LAN: A local Area Network (LAN) is a network that connects computer and other devices within a limited geographical area such as home, school or laboratory.

It allows devices to communicate with each other & share resource.

PROCEDURE:

(i) connect 4 computers to a switch using ethernet cables

(ii) connect 4 computers to a switch using ethernet cables

(ii) Assigned IP addresses:

⇒ PC1 - 10.1.1.1

⇒ PC2 - 10.1.1.2

⇒ PC3 - 10.1.1.3

⇒ PC4 - 10.1.1.4

⇒ Subnet mask - 255.0.0.0

(iii) Configure ~~connectivity using~~ the ping ~~command~~. Switch with IP addres 10.1.1.5

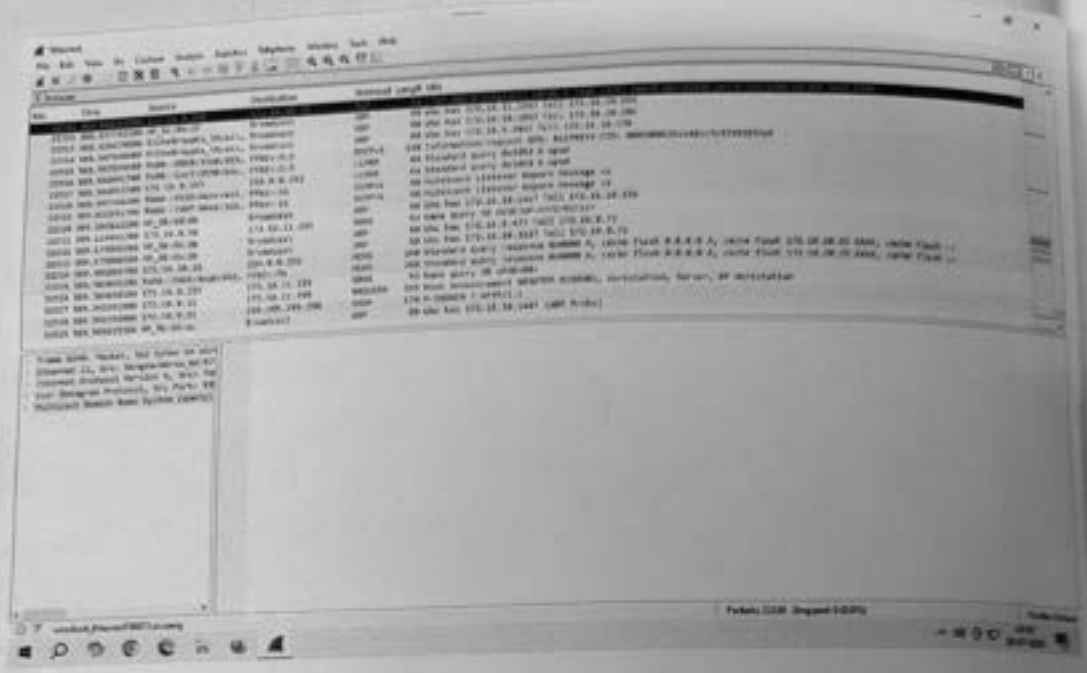
(iv) Verify connectivity using the ping command

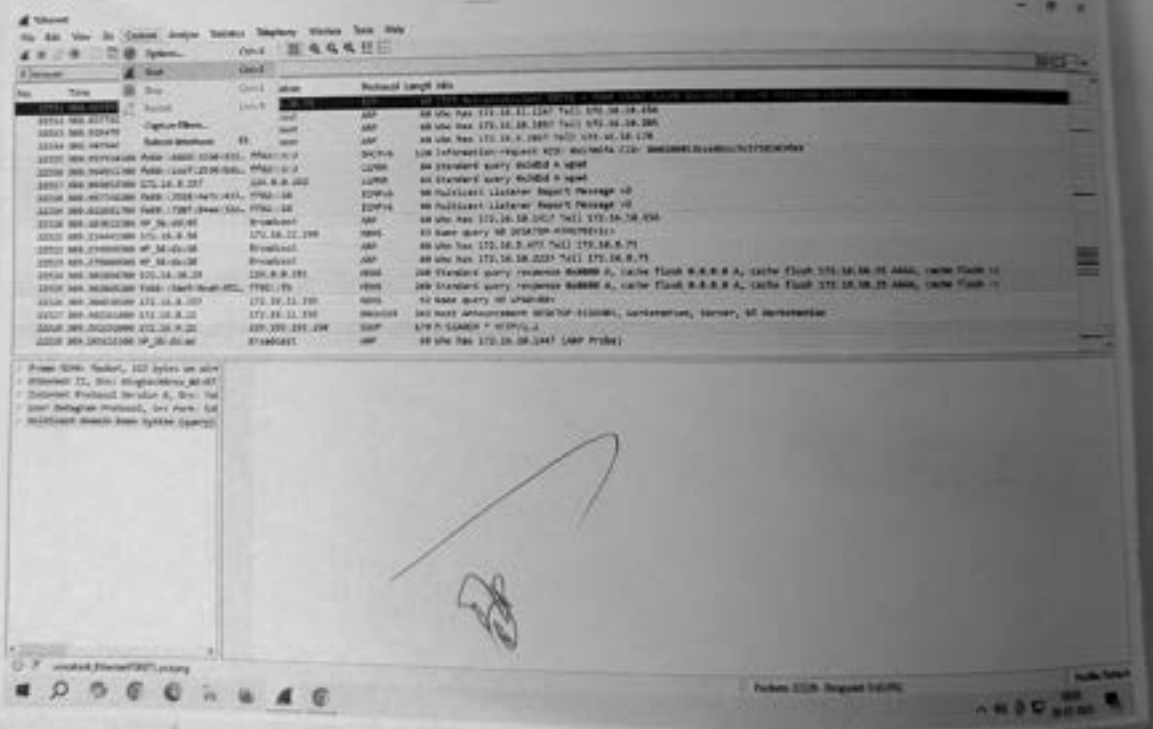
(v) Share a folder and accused it from the other PC's on the LAN

IP addres table

Device	IP address	Subnet mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0
Switch	10.1.1.5	255.0.0.0

Result: A local Area Network (LAN) was successfully set up and configured using a switch and ethernet cable. Connectivity between all devices was verified using ping command.





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6/8/26 PACK

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EXP NO: 5
6/8/26 PACKET CAPTURE USING WIRESHARK

AIM:

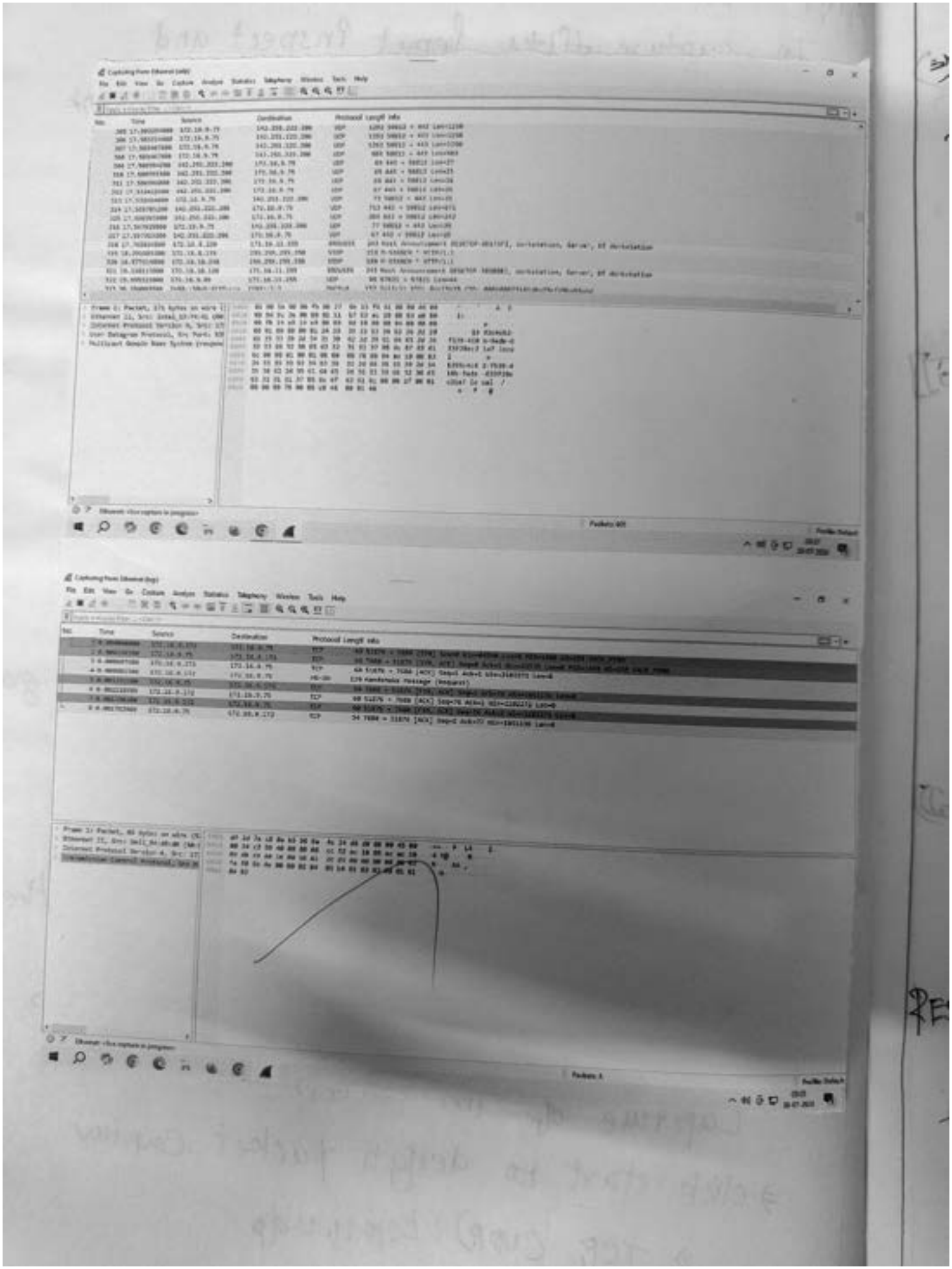
To capture filter Input Inspect and analyze network packets using the wireshark packet analyzer.

ALGORITHM:

1) open wireshark
2) select the active network Interface
3) Configure capture to stop after 100 packets
4) start packet capture
5) Apply display filters
6) Inspect packet details and view the flow graph
7) Save the captured Packets

PROCEDURE:

⇒ Launch wireshark and select the active network Interface
⇒ Go to capture → options and set stop Capture of 100 packets
⇒ click start to design packet Capture
⇒ TCP (UDP): tcp || udp
⇒ ARP : arp
⇒ DHCP : bootp
⇒ DNS : dns
⇒ HTTP: http



» Inspect the filtered packets in the packet list, details and bytes panes

» Save the capture file for future analysis.

RESULT: Successfully captured 100 Network packets, applied display filters for TCP, UDP, ARP, DNS, HTTP, IP, ICMP, DTCP analyzed packet details using Wireshark.

Exp No:6
8/8/26

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Code:

```
data = list (map (int, input ("Enter 4 bits:
                  split()))

p1 = data[0]^data[1]^data[3]
p2 = data[0]^data[2]^data[3]
p4 = data[1]^data[2]^data[3]

h = [p1, p2, data[0], p4, data[1], data[2], data[3]]
print ("Hamming code:", "".join(map(str, h)))
pos = int(input("enter error pos (1-7) 0 for
                no error:"))

if pos:
    h[pos-1] = 1
print ("Received code:", "".join(map(str, h)))
c1 = h[0]^h[2]^h[4]^h[6]
c2 = h[1]^h[2]^h[5]^h[6]
c4 = h[3]^h[4]^h[5]^h[6]
error = c1 + 2*c2 + 4*c4
if error:
    print ("error at pos:", error)
    h[error-1] = 1
    print ("corrected code:", "".
join(map(str, h)))
```

else: print ("No error")
print ("recovered Data: ", h[2], h[4],
h[5], h[6])

Sample Input: 1 0 1 1

3

Sample output:

Hamming code: 0110011

Received code: 0100011

Error at position: 3

Corrected code: 0110011

Recovered Data: 1011

Exp No:6 Error Detection and Correction using Hamming code
8/8/26

AIM:
To write a program to Implement error detection and correction using the hamming code.

Algorithm:

* Start the program.
* Read the number of 4 bit data blocks (n) forming the input data stream
* For each bit data block, D = d_1 d_2 d_3 d_4, perform Hamming encoding

(i) Arrange 7 bits with parity positions 1, 2, 4

(ii) Compute the parity bits using even parity

(iii) Form the 7-bit Hamming code word. P_1 P_2 d_1 P_4 d_2 d_3 d_4

(iv) Compute error position: C_4 C_2 C_1

(v) If error position = 0, no error has occured else, the bit at "error position" is inverted -

vi) stop the program

RESULT:

The program for error detection and correction using Hamming Code was implemented sucessfully.

Exp No 7

8/8/26 flow control at data link layer using sliding window protocol.

AIM:

To implement flow control at the data link layer using the sliding window protocol and to simulate the flow of frames from one node to another.

ALGORITHM:

1. Start the program
2. Read total number of frames, window size and last frame number.
3. Send frames within the current window
4. If a frame is lost, retransmit from the lost frame.
5. Receive ACK's and slide the window.
6. Repeat until all frames are sent.
7. Stop

Sample I/P:

Enter total frames: 8

Enter window size: 4

Enter lost frame (-1 for None): 5

Sample O/P:

window 0 to 3

Sending Frame 0

Sending Frame 1

sending Frame 2

Sending frame 3

ACK 0

ACK 1

ACK 2

ACK 3

window : 4 to 7

Sending Frame 4

Sending Frame 5

Frame 5 lost

Retransmitting from Frame 5

Sending Frame 5

sending Frame 6

sending Frame 7

ACK 4

ACK 5

ACK 6

ACK 7 All frames transmitted successfully

code:

n= int

w=int

lost =

i=0

while

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Code:

```
n= int (input (" enter total frames: "))
w=int (input (" enter window size: "))
lost = int (input (" enter lost frame (-1 for None)"))
i=0
while i<n:
    print ("In window :", i," to", min(i+w-1, n-1))
    for j in range (i, min (i+w,n)):
        print (" sending Frame", j)
    if j == lost:
        print (" Frame", j, "LOST")
        print (" Retransmitting from Frame", j)
        for k in range (j, min (i+w, n)):
            print (" sending Frame", k)
        lost = -1
    For j in range (i, min (i+w, n)):
        print ("ACK", j)
    i+= w
print (" All frames transmitted Successfully")
```

RESULT: The program to implement flow control at data link layer using the sliding window protocols was successfully implemented.

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8a) Discover Live Hosts using Nmap

Aim:
To discover live hosts in a network using Nmap by performing ARP, ICMP, TCP and UDP based host discovery on TryHackMe Nmap room.

Procedure:

1. open the TryHackMe Nmap room and start the AttackBox
2. Identify the target IP address/subnet to be scanned.
3. Use Nmap host-discovery options to identify active systems.
4. perform ARP-based discovery using nmap -PR -sn Target
5. perform ICMP Echo discovery using -PE option and study ICMP Timestamp (-PP) and address mask (-PM) discovery.
6. perform TCP SYN and TCP ACK host discovery using -PS and -PA and study



8). configuration of wireless LAN using Cisco packet Tracer

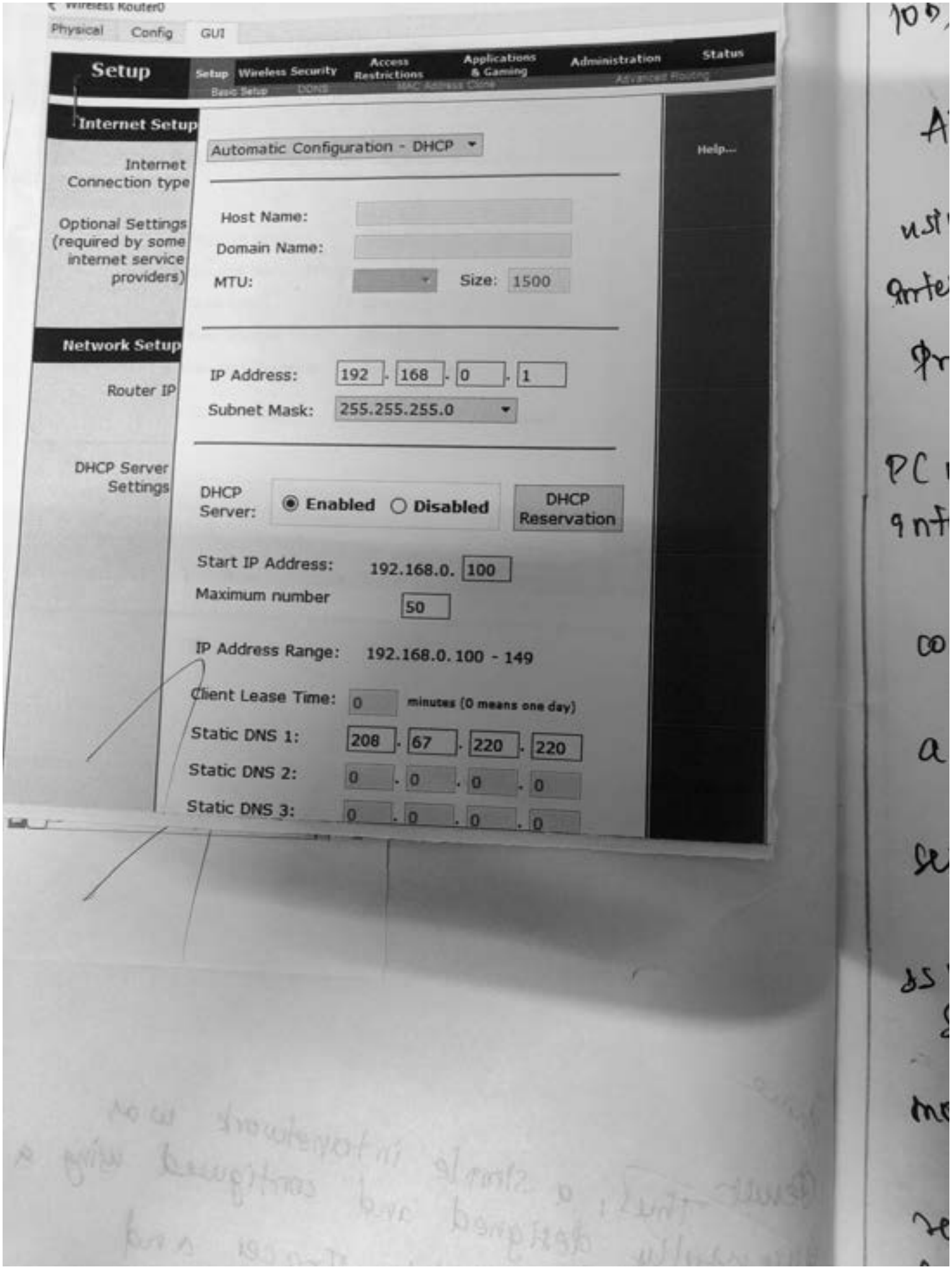
AIM: To design and Configure a wireless LAN using a Linksys- wireless router and three PCs in cisco packet Tracer with static IP addressing and WEP security.

procedure:

1. place one Linksys wireless Router and three PCs (PC0, PC1 and PC2) in packet tracer.
2. Configure the router SSID as MotherNetwork
3. Set the router IP address to 192.168.01 with subnet mask 255.255.255.0
4. Configure static IP addresses
   - PC0 - 192.168.0.2
   - PC1 - 192.168.0.3
   - PC2 - 192.168.0.4
6. Set the subnet mask to 255.255.255.0 and default gateway
7. open PC wireless, refresh the available networks and select motherNetwork.

3. Enter WEP key 0123456789 and connect each PC to the wireless router and verify wireless connection using ping

Result: The wireless LAN was successfully configured in cisco packet Tracer. The three PCs were connected to the MotherNetwork using static IP addresses and secured with configured WEP key


Wireless Router0
Physical
Config
GUI
Setup
Setup
Wireless Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Setup
DDNS
MAC Address Clone
Advanced Routing
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Help...
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500
Network Setup
Router IP
IP Address: 192 . 168 . 0 . 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address: 192.168.0. 100
Maximum number 50
IP Address Range: 192.168.0.100 - 149
Client Lease Time: 0 minutes (0 means one day)
Static DNS 1: 208 . 67 . 220 . 220
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0

10b). wireless Routers, DHCP Server and Internet cloud

Aim: To design and configure an Internetwork using a wireless router, DHCP server and Internet cloud in cisco packet Tracer

Procedure:

1) open cisco packet Tracer and place a PC, laptop, wireless router, cable modem, Internet cloud and cisco.com server.

2) Connect PC to wireless router using copper-straight through cable,

3) Connect wireless router to cable modem and cable modem to the Internet cloud.

4) Connect Internet cloud to cisco.com server using required cables.

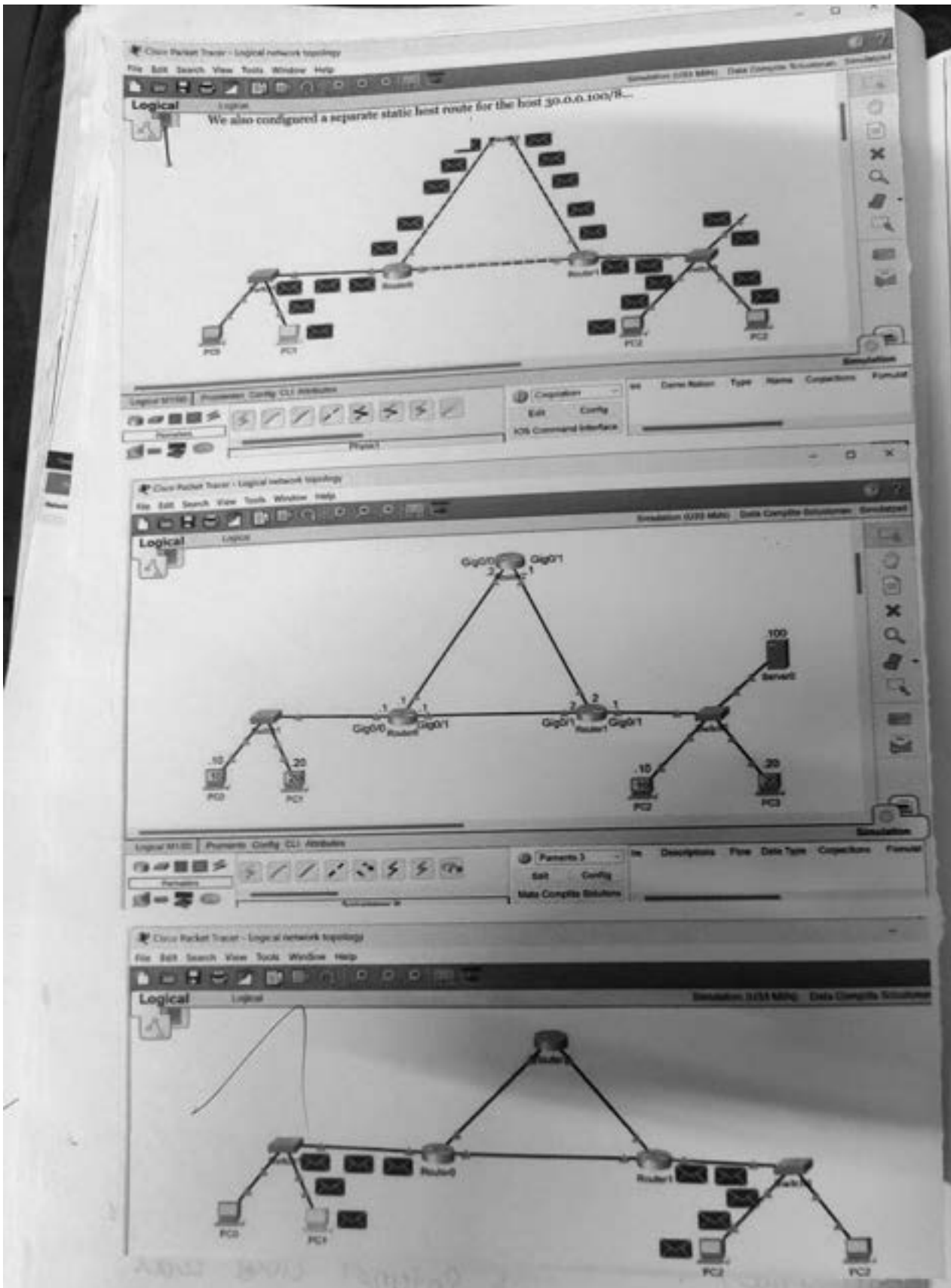
5) Configure the wireless router with SSID HomeNetwork and enable the DHCP server.

6) Configure laptop with wireless WPC300N module and connect it to HomeNetwork

7) use ipconfig/release and ipconfig/renew on PC to obtain network configuration from DHCP

8) Test connectivity by using the command ping cisco.com and verify received packets

Result: Thus, the Internetwork using a wireless router, DHCP servers and Internet cloud was successfully designed and configured, connectivity to cisco.com server was successfully verified



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11a) Simulate static Routing Configuration using cisco packet Tracer.

Aim: To simulate static Routing Configuration using Cisco packet Tracer and verify the connectivity between different networks using primary and backup route

Procedure:

1) open cisco packet Tracer and create required network topology with PCs and routers

2) Assign the given IP addresses and Subnet masks to all PC and router interfaces

3) Enable the router interfaces using no shutdown command.

4) Configure static route to remote networks using ip route command

5) Configure primary and backup route using different Administrative Distance values

6) Use show ip route static to verify the configured static routes

7) use ping and tracert commands to

test connectivity.

7. Disconnect the primary link between the routers to simulate a link failure.
8. Check the routing table again and observe that the back up route becomes active.
9. verify the connectivity again using ping and tracert after the link failure.

Result: Thus, static routing was successfully simulated using cisco packet Tracer. primary and backup routes were configured using Administrative distance. and the backup route was successfully used when primary route failed.

11) b) Routing Information Protocol

Aim:
To configure RIP routing using cisco packet tracer.

Procedure:

1. open cisco packet tracer and create the required router and PC topology.

2. connect the devices using suitable cables and assign IP addresses to PCs & router interfaces.

3. enable router R1 Interfaces using no shutdown and enter RIP configuration mode using router rip.

4. Add the connected networks using network commands.

5 Configure RIP on the required routers and check the routing table

6. Test the connectivity using 'Ping' ~~ping~~ command

Result: Thus RIP routing was successfully configured and connectivity between the networks was verified.



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12a) Echo client Server

Aim: To implement an echo client-Java using TCP/UDP sockets.

procedure:

1) Create the server socket and bind it to an IP address and port

2) start listening for client requests and Create the client socket

3) connect the client to the server and enter a message from the client

4) send the message to the server.

5) Server receives and echoes the message and client receives the echoed message

6) close the sockets.

Result: Thus, the echo client-Server was Sucessfully implemented using sockets.

6)b) chat client-server

Aim: To implement a chat program using socket programming

procedure:

1) Create the server socket and bind it to an IP address and port
2) start listening for a client and create the client socket.
3) Establish connection between client and server and enter a message from the client.
4) Send and receive messages between client and server.
5) Display the received message and continue communication until termination
6) close the sockets

Result: Thus the chat client-server was successfully implemented using socket programming

Python program:

```
import socket
import struct
import time
import os
def checksum(data):
    total=0
    for i in range (0, len(data)-1,2)
        total+=(data[i] << 8 + data[i+
    if len(data)%2:
        total+= data[-1] << 8
    total= (total >> 16)+ (total & 0xffff)
    total+ =total>>16
    return (~total) & 0xffff
def ping(host):
    try: dest_addr=socket.gethostbyname
        sock=socket.socket(socket.AF_
                socket.SOCK_RAW, Soc
    packet_id = os.getpid() & 0xffff
    sequence =1
    header =struct.pack("!BBHHH", ICM
            packet_id, sequence)
    data= b"PING"
    chksum = checksum(header+data)
    header =struct.pack("!BBHHH", ICM
                chksum, packet_id
                sequence)
    packet = header + data
```

B. Implement

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8. Implement your own ping program

Aim: To implement a simple ping program using socket programming to check whether a destination host is reachable and To calculate the response time.

Algorithm: 1) Start the program and get the destination IP address / hostname from user.

2) Create raw socket using Icmp protocol

3) Construct an Icmp echo request packet and record the time before sending the packet.

4) Send ICMP packet to the destination host and wait for Icmp echo reply.

5) Record the time when the reply is received

6) Calculate the Round Trip time (RTT) and display whether the host is reachable.

7) Repeat the process for required no. of packets and close the socket and Terminate the program.

Result: Thus, a ping program was implemented successfully using socket programming to determine host reachability and calculate round-trip time.

14. packet sniffing using Raw sockets

Aim: To implement a packet sniffing program using raw sockets and capture network packets to analyze their basic information such as source address, destination address and protocol.

Algorithm: ⇒ start the program and create a raw socket for capturing network packets

⇒ Bind the socket to the required network interface, if necessary

⇒ Receive packets continuously from network and extract the IP heads from each packet

⇒ Identify the source IP address and destination IP address

⇒ determine the protocol type such as TCP UDP or ICMP and display packet information

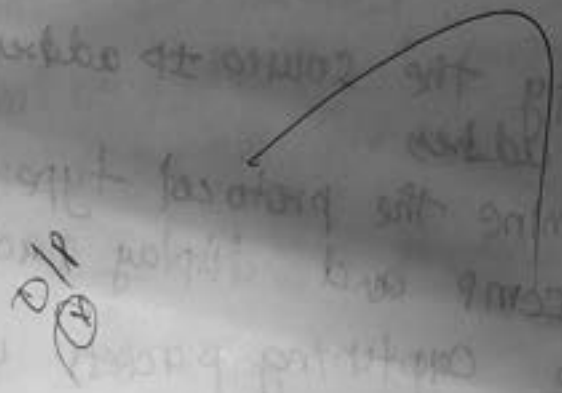
⇒ continue capturing packets until the user stops the program

⇒ close the raw socket and terminate the program

Result: Thus a packet sniffing program was implemented successfully using raw sockets, and the captured packets were analyzed based on their source IP, destination IP and protocol

I/P:
access key

O/P:
webalizer statistics:
Total requests : 1250
Total visits : 480
Total pages : 920
Total bytes : 15.4 MB
Succesful request : 1180
404 errors : 45



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⇒ open

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Result:

15. Analysis of various types of Services using webalizer

Aim:
To analyze the web server log files using the webalizer tool and study various server statistics such as no of requests, visitors, Transferred data, URLs and response status code

Algorithm: ⇒ start the webalizer application and install and configure webalizer on system
⇒ Locate the web server access log file and configure webalizer with required log-file path and output directory
⇒ Run webalizer on the selected log file
* Webalizer reads and analyzes the server log entries.
⇒ Generates statistics such as:
* No. of visits
* No. of requests
* pages accessed
* Data Transferred
* HTTP response codes
* Referring sites.
⇒ open the generated HTML report in web browser
⇒ Analyze the server activity and repeat the analysis for different log files.

Result: Thus the webalizer tool was used successfully to analyze web server log files, and various server statistics such as requests, bandwidth usage and HTTP response were obtained.